A Hybrid Temporal and Probabilistic Framework for Sepsis Prediction Using TA-BiLSTM, Dynamic Bayesian Networks, and Adaptive Particle Filtering

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***Abstract*—Sepsis is a life-threatening condition that requires early recognition of patient deterioration. Predicting sepsis from intensive care unit (ICU) time-series data is challenging because physiological measurements change over time and clinical observations may be incomplete.**

This paper presents a hybrid temporal and probabilistic framework integrating a Temporal Attention-based Bidirectional Long Short-Term Memory (TA-BiLSTM) encoder, a supervised Bayesian State Learner, a Dynamic Bayesian Network (DBN), and an Adaptive Particle Filter (APF). The framework processes clinical time-series data through preprocessing, sliding-window construction, temporal representation learning, state estimation, probabilistic temporal modeling, and sequential risk estimation. A classification threshold selected using validation data converts predicted probabilities into binary predictions.

The framework is evaluated using ROC-AUC, PR-AUC, accuracy, sensitivity, specificity, precision, and F1-score. Comparisons with conventional machine learning models and an LSTM baseline are presented using the supplied experimental metrics. The results are interpreted with attention to threshold selection, class imbalance, and consistency of the evaluation protocol.

***Index Terms*—Sepsis prediction, TA-BiLSTM, Bayesian state learning, Dynamic Bayesian Network, adaptive particle filtering, time-series analysis, ICU.**

I. Introduction

Sepsis is a serious clinical condition associated with organ dysfunction caused by a dysregulated response to infection. Early identification of patient deterioration is important because delays in recognition may lead to adverse outcomes.

ICU monitoring generates time-series data containing physiological measurements and laboratory observations. These data can provide useful information about changes in patient condition. However, clinical time-series prediction is complicated by missing measurements, irregular observations, temporal dependencies, and imbalanced outcome distributions.

Conventional machine learning methods can identify relationships between clinical features and outcomes. Recurrent neural networks, particularly Long Short-Term Memory (LSTM) networks, can learn temporal dependencies from sequential observations. Bidirectional LSTM networks process sequences in forward and reverse directions, while attention mechanisms can emphasize informative representations.

Temporal feature extraction alone does not explicitly represent the evolution of hidden clinical states. Probabilistic models offer a complementary approach by representing state transitions and uncertainty. Dynamic Bayesian Networks describe temporal dependencies, while particle filtering estimates evolving state distributions using weighted particles.

This study investigates a hybrid framework combining temporal feature extraction and probabilistic inference for sepsis risk prediction.

A. Research Objectives

The main objectives are:

1) To preprocess ICU time-series data and construct sliding-window sequences.
2) To extract temporal representations using a TA-BiLSTM encoder.
3) To learn Bayesian state representations from temporal embeddings.
4) To model state transitions using a Dynamic Bayesian Network.
5) To estimate sequential posterior risk using an Adaptive Particle Filter.
6) To compare the framework with baseline machine learning and recurrent neural network models.

B. Main Contributions

The framework investigates the integration of:

- Temporal representation learning using a TA-BiLSTM encoder.
- Supervised learning of discrete Bayesian states.
- Temporal probabilistic modeling through a Dynamic Bayesian Network.
- Sequential posterior estimation using an Adaptive Particle Filter.
- Threshold-based classification and comparative performance evaluation.

II. Related Work

A. Machine Learning for Sepsis Prediction

Traditional classifiers such as logistic regression, random forests, LightGBM, and XGBoost have been used for clinical

risk prediction. Their performance depends on the quality of the input features, data preprocessing, class distribution, and evaluation protocol.

B. Temporal Deep Learning

LSTM networks use gating mechanisms to capture temporal dependencies. Bidirectional architectures combine forward and backward contextual information. Attention mechanisms can further emphasize relevant features or time steps.

For prospective clinical prediction, the input sequence must contain only observations available at the prediction time.

C. Probabilistic Temporal Models

Dynamic Bayesian Networks represent dependencies among variables across time. Particle filtering provides a sequential approximation to posterior state distributions when analytical inference is difficult.

The proposed framework combines these ideas to investigate temporal representation learning and probabilistic state estimation for sepsis prediction.

III. Problem Formulation

Let the clinical feature vector at time t be

$$\mathbf{x}_t \in \mathbb{R}^d, \tag{1}$$

where d represents the number of input features.

A sliding window of length L is defined as

$$\mathbf{X}_t = [\mathbf{x}_{t-L+1}, \ldots, \mathbf{x}_t]. \tag{2}$$

The binary outcome is represented by

$$y_t \in \{0, 1\}, \tag{3}$$

where the label definition follows the dataset annotation protocol.

The objective is to estimate the probability of sepsis from observations available up to time t:

$$\hat{p}_t = P(y_t = 1 \mid \mathbf{X}_{1:t}). \tag{4}$$

A decision threshold τ converts the estimated probability into a binary prediction:

$$\hat{y}_t = \begin{cases} 1, & \hat{p}_t \geq \tau, \\ 0, & \hat{p}_t < \tau. \end{cases} \tag{5}$$

The threshold is selected using validation data and fixed before evaluating the held-out test set.

IV. Proposed Methodology

A. Overall Architecture

The proposed framework consists of the following stages:

1) ICU time-series data collection.
2) Missing-value treatment and normalization.
3) Construction of 24-hour sliding windows.
4) TA-BiLSTM temporal feature extraction.
5) Generation of 64-dimensional embeddings.
6) Supervised Bayesian state learning.
7) Four-state Dynamic Bayesian Network modeling.
8) Adaptive Particle Filtering.
9) Posterior sepsis probability estimation.
10) Validation-selected thresholding and evaluation.

The complete processing pipeline is illustrated in Fig. 1.

B. Data Preprocessing

Clinical observations are organized into time-ordered sequences. Each sequence is represented as

$$\mathbf{X}_i \in \mathbb{R}^{L \times d}, \tag{6}$$

where $L = 24$ denotes the sequence length in time steps and d denotes the number of input features.

Missing-value treatment and normalization are applied to prepare the input data. Data-dependent preprocessing parameters must be estimated using the training set and then applied unchanged to validation and test data.

Patient-level separation is important when constructing overlapping windows. Windows belonging to the same patient should not be distributed across training and test partitions when evaluating generalization to unseen patients.

C. TA-BiLSTM Temporal Encoder

The TA-BiLSTM encoder extracts temporal information from the input sequence.

The LSTM input gate is defined as

$$\mathbf{i}_t = \sigma(\mathbf{W}_i \mathbf{x}_t + \mathbf{U}_i \mathbf{h}_{t-1} + \mathbf{b}_i). \tag{7}$$

The forget gate is

$$\mathbf{f}_t = \sigma(\mathbf{W}_f \mathbf{x}_t + \mathbf{U}_f \mathbf{h}_{t-1} + \mathbf{b}_f). \tag{8}$$

The output gate and candidate cell state are

$$\mathbf{o}_t = \sigma(\mathbf{W}_o \mathbf{x}_t + \mathbf{U}_o \mathbf{h}_{t-1} + \mathbf{b}_o), \tag{9}$$

$$\tilde{\mathbf{c}}_t = \tanh(\mathbf{W}_c \mathbf{x}_t + \mathbf{U}_c \mathbf{h}_{t-1} + \mathbf{b}_c). \tag{10}$$

The cell state and hidden representation are

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t, \tag{11}$$

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t). \tag{12}$$

Here, σ denotes the sigmoid function and $\odot$ denotes element-wise multiplication.

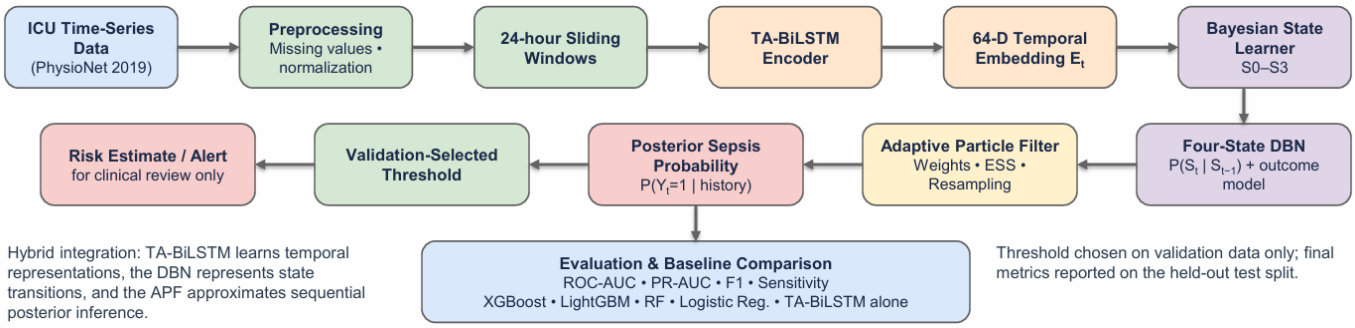


Fig. 1. Overall architecture of the proposed hybrid sepsis prediction framework, including clinical data preprocessing, sliding-window construction, TA-BiLSTM encoding, Bayesian state learning, DBN modeling, adaptive particle filtering, threshold selection, and performance evaluation.

The bidirectional representation combines forward and backward hidden states:

$$\mathbf{h}_t^{\mathrm{bi}} = [\overrightarrow{\mathbf{h}}_t; \overleftarrow{\mathbf{h}}_t]. \tag{13}$$

The encoder produces a temporal embedding

$$\mathbf{z}_t = f_\theta(\mathbf{X}_t), \tag{14}$$

where f_θ denotes the trained encoder. The proposed implementation uses a 64-dimensional embedding, as indicated in the framework diagram.

D. Supervised Bayesian State Learner

The Bayesian State Learner maps the temporal embedding to a discrete state representation. Let

$$s_t \in \{S_0, S_1, S_2, S_3\} \tag{15}$$

represent the four possible states.

The state learner estimates a mapping between the temporal embedding and the state representation, which can be expressed generally as

$$P(s_t \mid \mathbf{z}_t). \tag{16}$$

The learned states provide a compact representation for subsequent probabilistic temporal modeling. The exact state-learning objective and estimation procedure should follow the implemented algorithm.

E. Dynamic Bayesian Network

The Dynamic Bayesian Network represents dependencies between successive states and the observed outcome.

The state transition probability is

$$P(s_t \mid s_{t-1}), \tag{17}$$

and the outcome model can be expressed as

$$P(y_t \mid s_t). \tag{18}$$

A simplified first-order factorization is

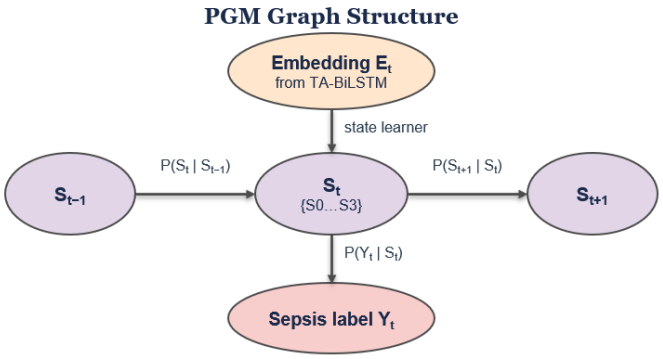


Fig. 2. Probabilistic graphical model showing the relationship between temporal embeddings, successive Bayesian states, and the sepsis label.

$$P(s_{1:T}, y_{1:T}) = P(s_1) \prod_{t=2}^{T} P(s_t \mid s_{t-1}) \prod_{t=1}^{T} P(y_t \mid s_t). \tag{19}$$

The probabilistic dependencies between temporal embeddings, successive states, and the sepsis outcome are illustrated in Fig. 2.

The graph provides a conceptual representation of the state-transition and outcome dependencies. Its conditional probability relationships should match the actual DBN implementation.

F. Adaptive Particle Filter

The Adaptive Particle Filter approximates a hidden state distribution using a collection of weighted particles:

$$\mathcal{P}_t = \{s_t^{(1)}, s_t^{(2)}, \ldots, s_t^{(N)}\}. \tag{20}$$

The particle weights satisfy

$$\sum_{j=1}^{N} w_t^{(j)} = 1. \tag{21}$$

Particles are propagated using the state transition model:

$$s_t^{(j)} \sim P(s_t \mid s_{t-1}^{(j)}). \tag{22}$$

TABLE I
DATASET SUMMARY

Property	Value
Original dataset	PhysioNet/CinC 2019
Original patient records	40,336 (verify release)
Input features in pipeline	40
Sequence length	24 time steps
Training sequences	52,366
Validation sequences	6,050
Test sequences	6,051
Total processed sequences	64,467

The observation likelihood updates the unnormalized weights:

$$\tilde{w}_t^{(j)} = w_{t-1}^{(j)} P(\mathbf{z}_t \mid s_t^{(j)}). \tag{23}$$

The normalized weights are calculated as

$$w_t^{(j)} = \frac{\tilde{w}_t^{(j)}}{\sum_{k=1}^{N} \tilde{w}_t^{(k)}}. \tag{24}$$

The effective sample size is

$$N_{\text{eff}} = \frac{1}{\sum_{j=1}^{N} (w_t^{(j)})^2}. \tag{25}$$

Resampling can be performed when the effective sample size falls below a predefined threshold. The actual particle count, adaptation rule, resampling method, and likelihood function must be specified according to the implementation.

Ground-truth test labels must not be used to update particle weights during inference.

G. Posterior Risk and Threshold Selection

The posterior sepsis probability can be represented generally as

$$\hat{p}_t = P(y_t = 1 \mid \mathbf{z}_{1:t}). \tag{26}$$

This probability is converted to a binary prediction using a threshold selected on validation data.

$$\hat{y}_t = \mathbb{I}(\hat{p}_t \geq \tau), \tag{27}$$

where $\mathbb{I}(\cdot)$ denotes the indicator function and τ is the selected threshold.

The threshold is fixed before final test evaluation. The resulting risk estimate is intended for research evaluation and clinical review, not as a standalone clinical diagnosis.

V. EXPERIMENTAL SETUP

A. Dataset Summary

The source dataset used in this project is identified as the PhysioNet/Computing in Cardiology Challenge 2019 dataset. The original-record count and the processed-window counts describe different quantities and should be verified against the exact downloaded release and split-generation scripts.

The training, validation, and test counts above are processed sequences, not unique patients. Because sliding windows can overlap, these counts must not be reported as patient counts. Confirm the original dataset release and verify that the split protocol prevents leakage before publication.

B. Evaluation Metrics

Accuracy is defined as

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}. \tag{28}$$

Sensitivity is

$$\text{Sensitivity} = \frac{TP}{TP + FN}. \tag{29}$$

Specificity is

$$\text{Specificity} = \frac{TN}{TN + FP}. \tag{30}$$

Precision is

$$\text{Precision} = \frac{TP}{TP + FP}. \tag{31}$$

The F1-score is

$$F_1 = 2\frac{\text{Precision} \times \text{Sensitivity}}{\text{Precision} + \text{Sensitivity}}. \tag{32}$$

ROC-AUC measures discrimination across classification thresholds, while PR-AUC summarizes the precision-recall trade-off. Both metrics complement threshold-dependent measures such as accuracy and F1-score.

VI. RESULTS

All baseline metrics in Table II are taken from the supplied test-set metric files. The proposed model row uses the supplied optimized metrics at threshold $\tau = 0.13$. Because threshold optimization can affect threshold-dependent metrics, the threshold must be selected on validation data and the final row must be confirmed using held-out test predictions before publication.

As shown in Table II, the proposed model has the highest reported F1-score, accuracy, sensitivity, specificity, and precision among the displayed rows. LightGBM and XGBoost have higher ROC-AUC and PR-AUC values, indicating stronger overall ranking performance in those metrics. These results illustrate a trade-off between threshold-independent discrimination and threshold-dependent classification performance.

A. Performance Visualization

Figure 3 compares F1-score and sensitivity for the proposed model at thresholds 0.13 and 0.50 and for the baseline models.

The chart shows that the choice of classification threshold changes the balance between F1-score and sensitivity. The threshold must be selected on validation data only, then held fixed for final test evaluation. The two threshold points should be described as separate operating points rather than as separate models.

TABLE II
TEST-SET PERFORMANCE COMPARISON OF BASELINE MODELS AND THE PROPOSED BAYESIAN MODEL

Model	ROC-AUC	PR-AUC	F1-score	Accuracy	Sensitivity	Specificity	Precision
LightGBM	0.9314	0.7476	0.6680	94.60%	74.77%	96.15%	60.37%
XGBoost	0.9317	0.7500	0.6591	94.56%	72.27%	96.31%	60.57%
Random Forest	0.9291	0.6923	0.6520	95.31%	60.45%	98.04%	70.74%
Logistic Regression	0.7997	0.3408	0.3042	76.86%	69.55%	77.44%	19.47%
LSTM	0.7857	0.3793	0.3149	83.24%	52.95%	85.62%	22.40%
TA-BiLSTM (alone)	0.8289	0.4127	0.3328	80.12%	68.18%	81.06%	22.01%
Proposed Bayesian Model ($\tau = 0.13$)	**0.9054**	**0.5344**	**0.7568**	**96.94%**	**77.82%**	**98.18%**	**73.67%**

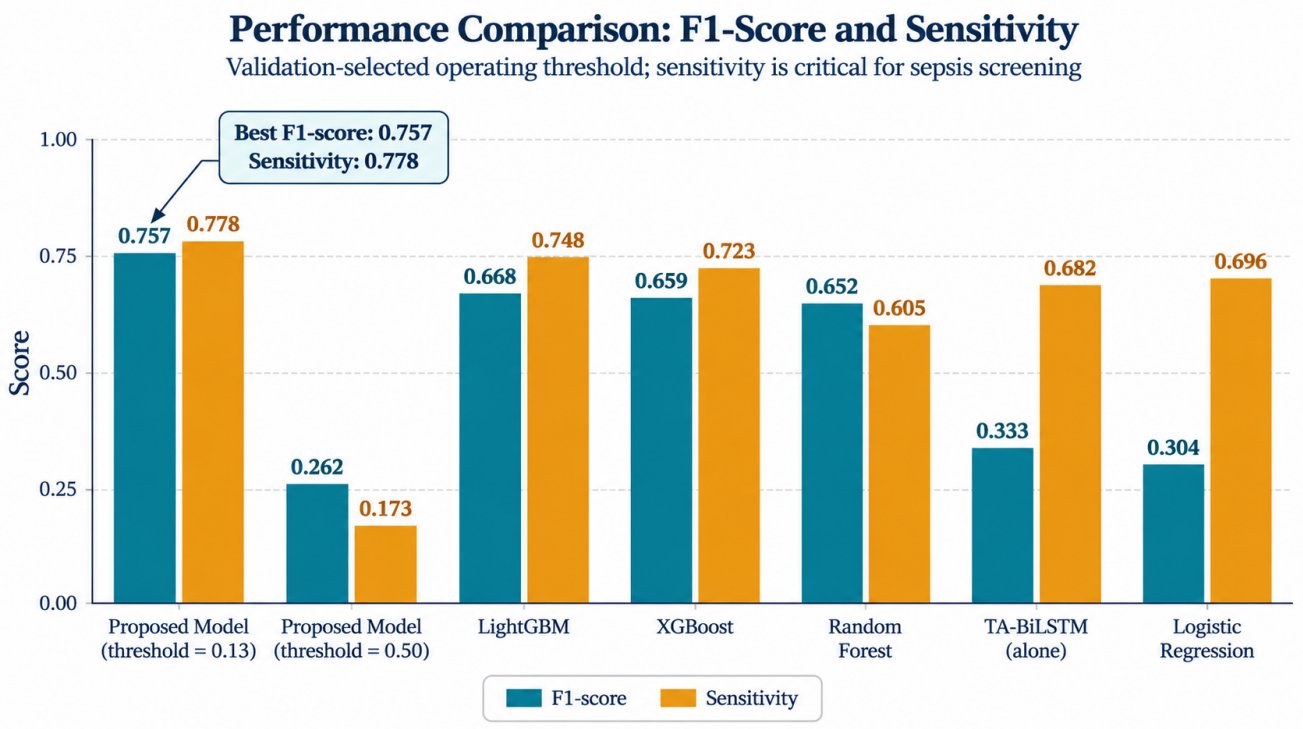


Fig. 3. F1-score and sensitivity comparison for the proposed model at thresholds 0.13 and 0.50, TA-BiLSTM alone, and baseline models.

VII. EXPERIMENTAL PROTOCOL AND REPRODUCIBILITY

A. Data Partitioning and Leakage Prevention

A reliable clinical prediction experiment depends not only on the choice of model but also on how observations are partitioned. The dataset used in this project is identified as the PhysioNet/Computing in Cardiology Challenge 2019 sepsis dataset. The processed files described in the project contain time-series features and outcome labels; they should not be confused with the original patient records. In particular, the number of generated windows is not the number of unique patients. A single patient sequence can generate many overlapping windows, and those windows are statistically dependent.

Where the intended use case is prediction for a previously unseen patient, partitioning should be performed at patient level before sliding windows are generated. If windows from the same patient appear in both training and test partitions, a model may benefit from patient-specific patterns that are unavailable for genuinely new patients. This can lead to overly optimistic estimates of generalization. The patient identifiers used to create each partition should therefore be checked for overlap, and the partitioning procedure should be documented in the final version of the paper.

All preprocessing operations that learn parameters from data must follow the same separation principle. For example, means and standard deviations used for normalization, imputation statistics, feature selection, and any learned discretization boundaries should be estimated from the training partition only. The resulting parameters are then applied to validation and test records without refitting. This rule prevents information from the evaluation data from influencing model development.

B. Training, Validation, and Test Roles

The three data partitions serve distinct purposes. Training data are used to estimate model parameters. Validation data support model selection, hyperparameter decisions, and the

selection of a classification threshold. The held-out test set is reserved for the final evaluation after these decisions have been frozen. Repeatedly adjusting the model or threshold after inspecting test performance weakens the independence of the reported test estimate.

This distinction is particularly important when comparing a threshold of 0.13 with a threshold of 0.50. A lower threshold can classify more observations as positive, which may increase sensitivity but can also increase false positives. The threshold should be selected according to a prespecified validation criterion, such as maximizing validation F1-score or meeting a minimum sensitivity requirement. The chosen criterion must be stated explicitly; it should not be inferred from the test results after the experiment has finished.

The reported optimized metrics list a threshold of 0.13. Before publication, the authors should confirm that this threshold was selected using validation data and that all threshold-dependent values in the proposed-model row were then recalculated from held-out test predictions. The adaptive-particle-filter metrics file also contains a different F1-score and sensitivity from the optimized-metrics file. These values should be reconciled by checking the exact prediction file, threshold, and evaluation partition used for each run. This reconciliation is necessary to ensure that the table represents one clearly defined experiment rather than a mixture of operating points or runs.

C. Reproducible Evaluation Procedure

A reproducible evaluation should save the prediction probability for every test observation together with its true label, patient identifier, and sequence or time index. These outputs make it possible to recompute ROC-AUC, PR-AUC, and threshold-dependent metrics independently. The final artifact should also record the feature list, preprocessing configuration, sequence length, random seeds, model hyperparameters, software versions, and the exact test split. If a model is stochastic, repeated runs or uncertainty intervals can help determine whether small differences are stable.

The evaluation procedure can be summarized as follows:

1) Freeze the patient-level training, validation, and test partitions.
2) Fit preprocessing parameters and train model parameters on training data.
3) Select hyperparameters and the operating threshold using validation data only.
4) Freeze the model, preprocessing parameters, and threshold.
5) Generate predictions for the held-out test set without consulting test labels during inference.
6) Compute all reported metrics from the same saved test predictions and document the positive-label definition.

This protocol makes model comparisons easier to reproduce and helps distinguish improvements in ranking quality from improvements caused by a different threshold. It also provides a clear audit trail if a reported value must later be corrected.

VIII. Additional Results Interpretation

A. Threshold-Independent and Threshold-Dependent Metrics

ROC-AUC and PR-AUC assess how well a model ranks positive examples ahead of negative examples across possible thresholds. They do not, by themselves, identify a single operating point. Accuracy, sensitivity, specificity, precision, and F1-score are calculated after a threshold has been applied and therefore depend on the selected decision rule. A model may have strong ranking performance but a less favorable F1-score at a particular threshold, or it may achieve a high F1-score at one operating point without having the highest area under either curve.

This distinction is visible in the supplied comparison. LightGBM and XGBoost report higher ROC-AUC and PR-AUC than the proposed Bayesian model, while the proposed row reports higher threshold-dependent F1-score, accuracy, sensitivity, specificity, and precision. These findings should be presented as a trade-off rather than as evidence that one method is superior on every dimension. In particular, the reported values should be interpreted only after confirming that they come from comparable test partitions and consistent label definitions.

B. Clinical Interpretation of Sensitivity and Specificity

In an early-warning setting, sensitivity describes the fraction of sepsis-positive examples that are detected by the model. A false negative may correspond to a positive case that does not trigger an alert at the selected threshold. Specificity describes the fraction of negative examples correctly classified and is related to the frequency of false-positive alerts. A high sensitivity can be desirable when missed cases are costly, but an increase in sensitivity may also increase the number of false alarms. The acceptable trade-off depends on the clinical workflow and cannot be determined from aggregate metrics alone.

Precision is also important because it measures the proportion of predicted-positive examples that are actually positive under the dataset's label definition. In datasets with relatively few positive examples, precision can remain modest even when specificity and accuracy appear high. Consequently, accuracy should not be used as the sole basis for selecting a sepsis prediction model. PR-AUC, sensitivity, precision, and F1-score provide additional context, while calibration and decision-curve analyses would be useful extensions for assessing whether predicted probabilities support meaningful decisions.

C. Component-Wise Ablation Requirements

The architecture combines several components: a temporal encoder, a discrete state learner, a DBN, and a particle filter. A full-system comparison alone cannot establish how much each component contributes. A future ablation study should evaluate a set of configurations under the same data split and evaluation protocol. Useful configurations include a TA-BiLSTM-only model, a TA-BiLSTM followed by the state learner and DBN without particle filtering, and the complete

system with particle filtering. If feasible, the state learner and transition model should also be tested separately to identify whether gains arise from temporal representation learning, explicit state transitions, or sequential posterior estimation.

For a fair ablation, all variants should use the same training and test patients, and threshold selection should follow the same validation procedure. Results should report both area-based metrics and threshold-dependent metrics. If the particle filter is adaptive, the adaptation rule and resampling criterion should be fixed or tuned on validation data. Ablation results should be reported only after running these experiments; no missing ablation value should be inferred from the full model's score.

D. Computational and Deployment Considerations

The proposed pipeline has several sequential stages, so its practical cost depends on more than the TA-BiLSTM encoder. Preprocessing and sliding-window generation determine how many sequences are evaluated. The encoder performs temporal feature extraction, while state inference and particle propagation introduce additional computations. For a particle filter with N particles and T time steps, the propagation and weight-update cost generally grows with both the number of particles and the sequence length. The actual runtime should be measured on the target hardware rather than estimated from model architecture alone.

A deployment-oriented evaluation should report inference latency per window or per patient, memory consumption, particle count, and the frequency at which predictions are updated. It should also define how missing or delayed measurements are handled and what happens when a required feature is unavailable. Because ICU measurements can be irregular, the model's behavior under different missingness patterns should be evaluated explicitly. These details are important for translating an offline experiment into a dependable monitoring component.

The system should be considered a decision-support research prototype, not an autonomous diagnostic system. Any alert would require clinical interpretation, and a deployment study would need prospective evaluation, external validation, calibration analysis, monitoring for distribution shift, and appropriate governance. The current results do not establish clinical effectiveness or readiness for bedside use.

IX. Discussion

The proposed framework combines temporal representation learning with probabilistic inference. TA-BiLSTM is intended to extract temporal patterns from clinical sequences. The Bayesian State Learner transforms these representations into discrete states, while the DBN models their temporal dependencies. The Adaptive Particle Filter estimates evolving state beliefs using weighted particles.

The combination is intended to represent both temporal information and uncertainty. However, the effectiveness of this integration must be established experimentally through consistent baseline evaluation and ablation studies.

Accuracy alone may not adequately describe a model's usefulness for sepsis prediction. Sensitivity measures the proportion of positive cases detected, while specificity measures the proportion of negative cases correctly identified. ROC-AUC and PR-AUC provide complementary measures of discrimination.

A rigorous evaluation should prevent information leakage across patient splits and ensure that true test labels do not influence inference. The threshold must be selected on validation data before the final test evaluation.

X. Limitations

The results depend on the dataset, label definition, preprocessing choices, and evaluation protocol. Missing measurements and irregular sampling may influence temporal representations. The Bayesian state model depends on its state-learning method and transition structure, while particle filtering introduces computational costs.

The supplied metric files should be checked against the final saved predictions to ensure that every row uses the same held-out test split. In particular, threshold-dependent metrics for the proposed model must be computed at the threshold selected using validation data, without using test labels to tune the threshold.

Evaluation on a single dataset does not establish generalization to other hospitals or patient populations. External validation, calibration assessment, and clinical evaluation are necessary before considering real-world deployment.

XI. Conclusion

This paper presents a hybrid temporal and probabilistic framework for sepsis risk prediction using ICU time-series data. The pipeline combines sliding-window preprocessing, TA-BiLSTM temporal encoding, supervised Bayesian state learning, Dynamic Bayesian Network modeling, and Adaptive Particle Filtering.

The framework aims to combine learned temporal representations with sequential probabilistic state estimation. The provided results offer an initial comparison with conventional machine learning and recurrent neural network baselines. However, the evaluation protocol and implementation must be verified before definitive performance claims can be made.

Future work includes consistent baseline evaluation, component-wise ablation studies, probability calibration, uncertainty analysis, and external validation.

Acknowledgment

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